Adding Board to the Xilinx Vivado

Step 1 – Download the Board Files : RealDigital AUP-ZU3 Board Page

Step 2 –Copy the extracted board folder into: C:\Xilinx\Vivado\2023.1\data\boards\board_files\

Package nlfm_waveform_gen as a Custom IP

Step 1 – Open the IP Packager

In Vivado: → Tools → Create and Package New IP →Choose: Package your current project(NLFM)

-→Click Next.

: Generate the NLFM ROM

- Generate the .coe file from Python.

Step 2 : Creating the Block Design

IP Integrator → Create Block Design

Search for:

- Clock Wizard
- nlfm_waveform_gen
- ILA
- Constant

Step 3: Configure the ILA

Probe widths:

Probe	Width
probe0	16
probe1	16
probe2	1
probe3	1

Step 4 : Add Constant IPs

You need two Constant blocks.

Constant 1

Value = 0

Rename: const_rst

Constant 2

Value = 1

Rename : const_start

Sub Step 1:In the Sources window →Right-click {design_1.bd}

Select → Create HDL Wrapper

Choose →Let Vivado manage wrapper and auto-update

Click OK

You should now see : design_1_wrapper.v

Sub Step 2

design_1_wrapper.v :Set it as Top

Sub Step 3

Generate Output Products

Right click → design_1.bd

Choose →Generate Output Products

Leave → Global

Click → Generate

Sub Step 4

Generate Bitstream prerequisites

Run →Open Elaborated Design

If it succeeds → Run Synthesis → Run Implementation→ Generate Bitstream

Step 7:

1. Connect the AUP-ZU3 board via JTAG.
2. Power on the board.
3. In Vivado:

Open Hardware Manager
↓
Open Target
↓
Auto Connect
↓
Program Device